CSD 598 - Winter 2026

An intro to programming and statistical linear model diagnostics

Pertti Palo

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Last time: Exploration and inference example

Any questions? Any thoughts?

Intro to programming

- Obviously there is a lot to dig into here.
- A sufficient understanding does not require you to know too many details.
- Time allowing, we'll look at different programming concepts as we work through some ways of diagnosing the behaviour or statistical linear models.
- First, though, we'll have a look at your questions and possibly use some of the following slides to answer them.
  - We may also wander into Python or R to try things out.

If you are feeling couragous install these

- RStudio `https://posit.co/downloads/`
- Python `https://www.python.org/downloads/`
- Depending on your operating system some form of app store maybe the best bet for installing anything.
- If you are super adventurous, install uv (`https://docs.astral.sh/uv/getting-started/installation/`) and use Python that way. (It is actually a really good idea.)

Basic building blocks I

```
# These are lines comments. They are meant for humans to read
# and are ignored by the computer.

# Operators: Do stuff.
1+1

# Variables: Make the results stick around.
a = 1+1
b = 3

# Combinations thereof: What most programming actually is.
c = b/a

# Functions and their siblings: Package recipes for later use.
squareroot(2)
```

Basic building blocks II

- integer – aka whole number,
- floating point number – aka decimal number,
- character – a single character,
- string – a piece of text made of characters, very useful to us,
- lists which will usually contain one given type of variable, e.g. a list of numbers or a list of strings.
- and many more.

Basic tools of the trade

- There are many programming languages and they are suited for different tasks and also differ in terms such as readability.
- Most relevant to us are R, Python, Matlab, Praat scripts, LibreOffice calc/Excel macros.
- Code files are usually identified with a suffix that corresponds to the language being used: e.g. `.R` for R, `.py` for Python, `.praat` for Praat (although the last one is not standardised).
- Code can be written with essentially any text editor that will save plain text (no formatting frills attached).

Actual tools of the trade

- Writing code generally easier with an actual code editor or IDE (Integrated Development Environment):
  - RStudio
  - Jasp
  - VSCode (corporate and a bit scary) / VSCodium (probably doesn't snitch on you)
  - PyCharm (often costs money)
  - Matlab
- Version control in some way, preferably with git if you end up doing this for any length of time.
- Consider using an AI helper (like Cursor, Copilot Agent, or Cline), but be aware that the less common your task, the more likely they are to produce only garbage.
- Helpers optimised for actual code writing are probably better than general generative models.

Skills of the trade

- Programming style: Code is read more than written. Learn to write in a clear style.
- Knowledge of the tools: There will always be new ones. Dedicate some time to learning new stuff.
- General principles like modularity, writing code to be read, documenting what you do, ...

Model diagnosis

- Collinearity (linear correlation) of the independent/predicting variables is bad (mathematically).
  - It can usually be spotted fairly easily by calculating correlations among the predictors and/or making scatter plots like a pairs plot.
- Heteroscedasticity will mess more the efficiency of prediction than the accuracy of it.
- Outliers can mess prediction accuracy.
- Here a bit of data exploration and model diagnostic plots go a long way:
  - Neat explanation of diagnostic plots can be found here `https://library.virginia.edu/data/articles/diagnostic-plots`
  - Collinearity can be seen in for example a pairs plot or a correlation matrix (not to be confused with e.g. "Correlation of Fixed Effects"in ‘lmer’ output).

Some resources

- RStudio `https://posit.co/downloads/`
- Jasp `https://jasp-stats.org/`
- Python learning materials `https://wiki.python.org/moin/BeginnersGuide/NonProgrammers`
  - Especially the Interactive Courses are really good for learning programming in general.
- Praat scripting guide `https://praatscriptingtutorial.com/`